On portfolio optimization based on machine learning predictions

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Abstract: In the paper we present a complex analysis of approaches to portfolio optimization based on predictions made by machine learning (ML) algorithms. We compare several ML algorithms prediction accuracy based on unified sample of feature variables. In the paper, we focus on segment of information technology (IT) companies listed on NYSE and NASDAQ exchanges and data between September 2017 and September 2025. Based on the results of our analyses, we propose a portfolio optimization strategy, which combines the Markowitz mean-variance approach with ML predictions.

Keywords: quantitative finance, portfolio optimization, machine learning, stock price prediction, stock segment

1 Introduction

Development of asset prices on financial markets is a complex and non-linear time series, which is affected by a large number of various external factors [Yin et al., 2025]. Having an ability of approximately predicting course of the asset prices in the future is of great significance among the investors. Confident price forecasting can be used for devising informed investment strategy and capital optimization.

This paper studies possibilities of devising unified data pipeline, capable of processing stock price prediction and implementing it in optimizing capital allocation on a given historical time interval.

Among a number of studies, involving forecasting price of stocks, many of them tend to shift focus on a very limited number of stocks [Huang, 2024, Yin et al., 2025] or on main-stream stock indices [Krauss et al., 2017]. There is a lack of study conducted on a coherent group of stock assets. What is more, is the fact that these papers tend to shift focus solely on the prediction results and mechanics and do not employ them in further application analyses, which are potentially of a of great importance for real-world investment decision-making.

The paper focuses on a certain stock market segment. Among 500 largest companies, listed on stock exchanges in the USA, by market capitalization, in total 65 of them belong to the market segment **Information technology** (IT). The main aim is to explore and evaluate prediction power of selected machine learning (ML) frameworks in predicting next-day closing price in the selected IT segment.

Consequently, such acquired predictions will be used as a foundation of building investment portfolio consisting of stocks from the given segment.

In the conclusion, paper implements and analyzes performance of a unified pipeline employing data pre-processing of a unified set of features related to the stock market, day-to-day closing price prediction and lastly portfolio optimization. From the broader perspective, paper brings study of predictability of the whole market segment on the US stock market.

2 Features used predictions

The amount of variables, which are directly or indirectly generated together with day- to-day market running, is quite enormous, and one should consider subsetting them for ML purposes wisely.

In this subchapter, we aim to show and justify the process of our general feature selection, mostly based on a priori knowledge of processes within financial markets and the variables' most likely effect on stock markets.

For the sake of completeness, data were obtained from the API directly fetching data available on Yahoo Finance – available through the Python abstraction provided by the `yfinance` library [Yahoo, , Aroussi, 2025].

2.1 Stock market assets

As a possible predictors for modelling, other market assets might be valuable. In the paper, we employed a selected number of **stock indices** – an abstract concept accumulating information regarding various stocks into one unified asset. Indices are specifically designed to replicate price development of a given set

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of stocks. By choosing well designed indices – replicating broad scope of the market – model might acquire valuable predictor information.

In the study, indices **S&P 500** and **NASDAQ Composite** (NASDAQ) were used as predictors. S&P 500 reflects performance of the 500 largest US-traded stocks in terms of total market capitalization. Unlike S&P 500, NASDAQ reflects performance of technology and IT stocks traded in the USA, which might presumably be beneficial, taking into account the scope of the study.

Furthermore, one of the most fundamental part of the markets play **commodities**. This term represents mainly raw goods which are traded on the markets in large quantities.

Their fundamental role in the market comes from the fact that most of these assets play an essential part in further processing, consumption and industry. Matsumoto et al. [Matsumoto et al., 2023] suggest, that the commodity market play a vital part in the whole financial segment, which suggests, that including such assets in the predictor database might be sufficient.

Among large number of commodities, we chose to include data regarding **gold** and **crude oil**. Gold still remains one of the most prominent securities. Most modern currencies are dependent on a system, where gold acts as a security stabilizing currency movements and limiting the governments issuing money [Osita, 2025]. Oil plays a vital role in keeping the world in motion, and a solid majority of the world's activities depend on its presence and availability. This might imply that these dependencies on the global market would be projected into the price movement of the commodity.

Lastly, in order ro reflect the international competitivness of US economy towards other major economies and quantitative information about relations among major economies, which is most likely an important predictor for stock price movements we decided to include data from the **foreign exchange market** (FOREX) – fully decentralized market reflecting relative value of particular currency to the other currency.

By our focus on the US market, we chose to include currency pairs with the Amer- ican dollar (USD) in all pairs. In particular, included pairs, together with the USD, are the Euro, Japanese yen, British pound sterling, Swiss franc, Australian dollar, Canadian dollar, and New Zealand dollar.

2.2 Other market indicators

In order to track the performance of the US economy more precisely, it is wise to include information regarding national debt as possible features. For this purpose, we included data about **US Treasury bonds**. According to Kumar et al. [Kumar et al., 2021], bonds of various maturities are a valuable market predictor; that is why we selected wide range of bond maturities in the predictor database. We included 13-week maturity (short term) and 5-year long maturity.

Apart from that, we included **Volatility Index** (VIX), which is a quantitative metric reflecting investor sentiment and is a sort of gauge tracking state of the US economy in terms of volatility and risk.

3 Prediction methodologies

3.1 Rolling window

In order to properly process time series data into training, validation and test sets many approaches arise as suitable [Diebold, 2006]. For the purposes of this paper and the **next-day prediction** – where data from trading days $t-q, t-(q-1), \ldots, t-1$ (for the given q) are employed as predictors for the closing price on day t – elements of rolling window might be adopted.

In our approach, we define static time series interval, which serves as a hybrid training-validation interval. The interval is between T_1 and T_2 timepoints (with $T_2 - T_1 = M$). On this interval reside two frames of variables.

Multivariate $\mathbf{X} \in \mathbb{R}^{M \times F}$, where F denotes the number of features used, serves as an input variable. Included features in $\mathbf{X}$ are values of selected financial variables, shifted backwards up to $q = 9$ days. That means, that $\mathbf{X}[t]$ (the t-th row of $\mathbf{X}$) represents data available through days $t-1, t-2, \ldots, t-9$.

Secondly, univariate $Y \in \mathbb{R}^M$ accumulates values of the target variable (next-day closing price) to the respective rows of $\mathbf{X}$. That means, that $Y[t]$ (the t-th value in the Y) is the closing price value on day t.

Furthermore, we define parameter $1 \leq p < M$ (training window length parameter), which is the length of the training subwindow, meaning how many next-day predictions model would be trained on. Lastly, parameter w represents the forecasting horizon – the number of time units into the future for

which predictions are made (we define $w = 1$).

With respect to the adopted data preparation and concepts of rolling window methodology, it is possible to generalize process of train and validation sets split. Algorithm 1 shows what parts of the global $\mathbf{X}, Y$ are directly available at the ceratin step of prediction.

Algorithm 1 Rolling-window on (T_1, T_2) interval

```
# We assume w ← 1
# The prediction target day is t + p + 1
# We have available data up to day t + p
Set look-back window length p
Retrieve the feature variables' values into the X
for each stock in database do
    Retrieve closing prices into the target Y
    for t = T1 to T2 − (p + 1) do
        X_train ← X[t : t + p]
        Y_train ← Y[t : t + p]
        X_predict ← X[t + p + 1]
        # X_predict is the latest available data
        Train model f̂ on (X_train, Y_train)
        Generate prediction: f̂(X_predict)
    end for
end for
```

In this paper, time interval between September 15, 2017 and December 31, 2022 (1333 trading days) is used as training-validation interval and time interval between January 1, 2023 and January 16, 2026 (761 trading days) is used as test interval.

3.2 ML frameworks

In the paper, various ML algorithms were employed. Taking into account rather short-period forecast horizon, regularized variants of linear regression might be a suitable framework. Longer forecast periods tend to be non-linear and patterns in them are harder to generalize. That is why we chose to employ variants **Lasso** and **Ridge**.

The core property of these algorithms is their inability to be time-aware – meaning they cannot capture time dependencies by default. One of the most widely used frameworks capable of it is the **Autoregressive Integrated Moving Average** (ARIMA) model. These three model frameworks were used as prediction engine for all the prediction analyses in the paper.

4 Predictive analyses

This part presents obtained modelling results. For each of the employed frameworks, in-depth parameter tuning and evaluation on both training-validation and test sets are reported.

4.1 Evaluation metrics

Based on the methodology introduced through Algorithm 1, suppose we obtain prediction for the day $t+1$ (of certain stock) as $\hat{Y}_{t+1}$. In order to measure relative deviation from the actual Y_{t+1}, metric of **absolute percentage error** (APE) is used. Particular APE is given as:

$$\mathrm{APE}(\hat{Y}_{t+1}, Y_{t+1}) = \frac{|Y_{t+1} - \hat{Y}_{t+1}|}{Y_{t+1}}.$$

Further, the average value of the APE for inidividual stock, over the entire considered time interval, is evaluated (further referred to as **mean absolute percentage error** – MAPE)

Apart from that, we introduce a baseline benchmark to measure the model's significance. According to Fama [Fama, 1970], the asset price development corresponds to the random walk. In accordance with that, such naïve benchmark can be introduced, that it predicts the next-day price on day $t+1$ to be equal to the price on day t. With that, we quantify it as proportion of predictions, made by the non-naïve model, that are closer (in terms of APE) to the actual closing price than naïve predictions. Further, this will be referred to as **model dominance**.

4.2 Lasso regression

Lasso regularization model is explicitly defined using two parameters. Earlier introduced p (length of the training window) and regularization parameter λ, steering the penalty in optimization.

Figure 1 visualizes relationship of MAPE on the chosen values of parameters λ and p for the sample chosen stock Enphase Energy Inc. (Ticker ENPH).

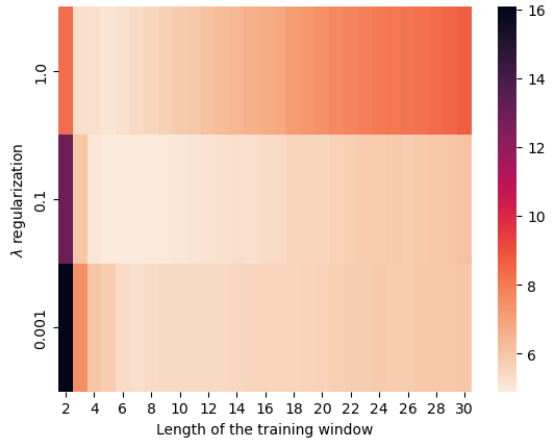

Figure 1: MAPE dependence on the rolling window training length parameter and regularization parameter (Lasso, stock ENPH).

In terms of the parameter p, it is possible to observe that the lowest MAPE is reported in the interval of $p \in [4,8]$. Based on the further analyses, we chose $p = 4$ being constant. With such value of the parameter p, the optimal (in terms of all stocks) interval for regularization parameter was localized being $\lambda \in [4,5]$. Figure 2 shows an overview of MAPE for the chosen ENPH stock in dependence form λ.

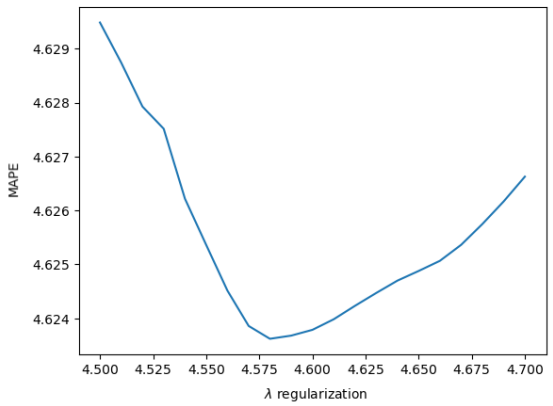

Figure 2: MAPE dependence on the rolling window training length parameter (Lasso, stock ENPH).

Such a convex course was observed in all the analyzed stocks, which means that localizing optimal value of λ was deterministic in all cases.

In evaluation of the Lasso framework we decided to implement two various approaches. Firstly, we set up the Lasso model with one universal value of $\overline{\lambda}$ being the arithmetic mean of all optimized values of λ of single stocks. The resulting value is $\overline{\lambda} = 4.15$ (referred to as General model). Secondly, model with variable value of λ, is based on identifying stock and applying its optimized value of λ in model (referred to as Stock-wise model).

Evaluation of these model on the training-validation interval is shown in Table 1:

Model Type	MAPE (%)	Dominance (%)
General model	2.297	36.16
Stock-wise model	2.27	35.81

Table 1: MAPE and dominance values for different settings of the model (Lasso).

What is interesting to point out, is the order of evaluation metrics for both models. It is possible to observe higher value of MAPE of General model; however, this setting achieved comparatively better dominance. Such behaviour could be considered counter-intuitive, as higher proportion of predictions being closer to the actual value of target variable might yield better overall deviation in general.

4.3 Ridge regression

With having the same model parameters as in the case of Lasso regression, it is possible to employ the same methodology of hyper-tuning the parameters.

Figure 3 shows the compound dependence of MAPE on the rolling window parameter p and regularization parameter λ (stock ENPH). Likewise the Lasso, this course in general applied to the vast majority of the analyzed stocks.

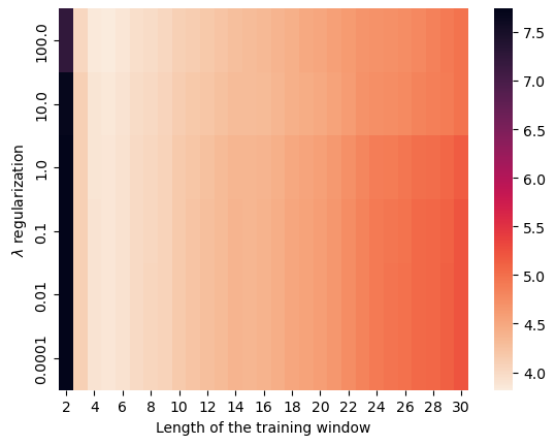

Figure 3: MAPE dependence on the rolling window training length parameter and regularization parameter (Ridge, stock ENPH).

In direct comparison with the Figure 1, it is possible to observe clear segment of near-optimal values of $p \in [4,6]$. Based on the further experiments of fitting the model, we chose the value $p = 4$.

Further, it is possible to analyze dependence of MAPE solely on λ regularization parameter. Figure 4 shows again convex course for the stock ENPH.

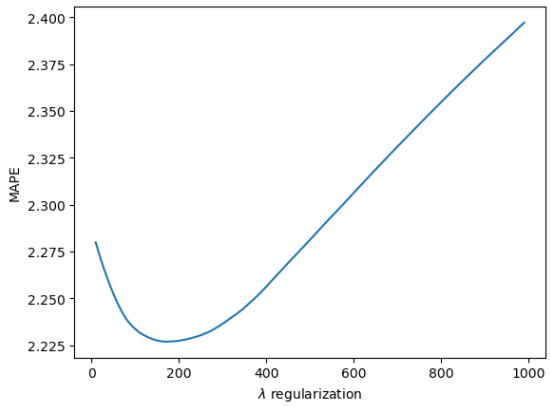

Figure 4: MAPE dependence on the rolling window training length parameter (Ridge, stock ENPH).

Even in this case, such course was observed in all of the analyzed stocks, therefore localizing the optimal value for each stock was determinsitic.

Lastly, in terms of model evaluation, we employed the same approach as in the Lasso framework. The average value of optimal λ's of each of the analyzed stocks if $\overline{\lambda} = 211.85$ (General model). The approach of using the optimal value of λ for each stock was preserved as well (Stock-wise model). Evaluation results are given in Table 2

Model Type	MAPE (%)	Dominance (%)
General model	1.791	45.08
Stock-wise model	1.79	45.15

Table 2: MAPE and dominance values for different settings of the model (Ridge).

Compared to the results in Table 1, order of models is preserved – Stock-wise model is superior in both of the metrics. What is more, relative differences of metric values between models is smaller, compared to the Lasso framework.

In terms of direct comparison, we observe marginal improvement compared to Lasso. Even though in terms of dominance, framework is inferior to the naïve benchmark, employing it showed progress.

4.4 ARIMA model

In the case of ARIMA we chose different approach to hypertune the parameters than in the previous two frameworks. For each particular rolling window, initial state $p = 0, d = 0, q = 0$ is set. Incrementally, algorithm explores the universe of possible settings by exploring the *neighboring* states in the grid. All possible combinations of integer settings are fitted on the given training data and Bayesian information criterion (BIC) is evaluated [Smith, 2025].

Subsequently, the optimal setting based on BIC is chosen. This method is somewhat naïve optimizer and, in terms of time complexity, com- putationally cumbersome. However, for the purposes of this research, this approach is suitable.

Therefore, in the case of ARIMA framework, there is no need to visualizes and report any particular dependencies as hypertuning is done ad hoc in every interval across all analyzed stocks.

Finally, in terms of evaluation, such set model achieved MAPE per stock **1.897**% and dominance of **58.25**%. ARIMA was able to overcome all of the analyzed frameworks (even naïve) in terms of model dominance. On the other hand, in terms of MAPE across stocks, Ridge model achieved better performance. Such result is particularly interesting, as a comparatively lower MAPE does not necessarily imply higher dominance.

4.5 Test evaluation

Last step in the complex predictive analysis is evaluating performance of optimized models on before unseen (test) data sets. Evaluation results for all the considered model variants is in Table 3

Model Type	MAPE (%)	Dominance (%)
ARIMA	1.923	56.68
Ridge (Stock-wise mode)	1.794	45.94
Ridge (General model)	1.798	45.88
Lasso (General model)	2.24	38.83
Lasso (Stock-wise model)	2.31	37.1

Table 3: Evaluation of tuned models on test interval.

Performance of models tend to reach similar values as on the training-validation interval – possibly no overfitting. Similar values of metrics indicate that

models perform relatively stable on both (differently wide) interval of data.

5 Portfolio management

With obtained models, capable of producing trustworthy predictions on the analyzed universe of stocks, next step is employing these models in process of portfolio optimization. In this paper, portfolio is meant as decision-making yielding allocation of capital across particular stocks in order to achieve various objectives (e.g., maximum return, minimal risk, etc.).

5.1 Theoretical foundations

Allocating capital to a given set of stocks in order to balance the trade-off of asset volatility and portfolio return can be formulated as an optimization problem. There are several approaches to properly constructing a portfolio strategy, each requiring different types of information from the data [Palomar, 2025].

Taking into account having prediction of returns for the next day as a suitable baseline, **Markowitz mean-variance portfolio** (MVP) was chosen. However, its elementary form tends to be impractical in practice [Palomar, 2025].

Its augmented form, introducing transactions costs was adopted. The complete definition of the used framework is as follows:

$$\max_{\mathbf{w}_t} \quad \mathbf{w}_t^\top \boldsymbol{\mu}_t - \lambda_{\mathrm{R}} \mathbf{w}_t^\top \boldsymbol{\Sigma} \mathbf{w}_t - \lambda_T \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1$$

$$\text{Subject to:} \quad \mathbf{w}_t \geq \mathbf{0}_a \text{ and } \mathbf{1}_a^\top \mathbf{w} = 1,$$

where variable $\mathbf{w}_t, \mathbf{w}_{t-1} \in \mathbb{R}^a$, where a is the total number of assets, are vectors of weights of particular stock assets in portfolio on days t and $t-1$ respectively. Further, $\boldsymbol{\Sigma} \in \mathbb{R}^{a \times a}$ is a covariance matrix of assets, λ_R is the risk-aversion hyperparameter and λ_T is the unit transaction cost parameter. Lastly, for $\mathbf{x} \in \mathbb{R}^t$, $\|\mathbf{x}\|_1 = \sum_{i=1}^t \|x_i\|$ is the L1 or Manhattan norm.

Such extended MVP framework not only balances return and variance of the possible portfolio, but penalizes high volume of transactions, which in practice imply lower losses [Palomar, 2025].

This allocation framework is used for daily reallocations, which means that in every trading day optimization is run and weights of single stock assets vary on daily basis.

Subsequently, it is necessary to define crucial evaluation metrics. Using them, comparison of portfolios between each other would be possible and interpretable.

Suppose portfolio is applied on time interval $[1, N]$ and value of the portfolio is a time series $V = V_1, V_2, \ldots, V_N$

Cumulative annual growth rate, representing the average percentage return per year, is given as follows [Palomar, 2025]:

$$\mathrm{CAGR}(V) = \left(\frac{V_N}{V_1}\right)^{\frac{1}{r}} - 1, \tag{1}$$

where r denotes number of trading years between 1 and N timepoints.

Further, to quantify risk and volatility of the portfolio **annualized volatility** (AV) can be adopted [Palomar, 2025]. Let $r_t = \frac{V_t}{V_{t-1}} - 1$ denote return on day t. Volatility across the given time interval is given as standard deviation of σ of $r_1, r_2, \ldots, r_N$. Subsequently, annualized volatility is given as follows:

$$\mathrm{AV}(V) = \sigma\sqrt{P}, \tag{2}$$

where P is the number of trading days per year.

Lastly, one compound metric, based on Equations (1) and (2) can be derived [Palomar, 2025]. **Sharpe ratio** stands for return strategy earns per unit risk and is given as:

$$\text{Sharpe ratio}(V) = \frac{CAGR(V)}{AV(V)} \tag{3}$$

5.2 Optimization on training-validation interval

With obtained predictions, the only parameter of the optimization model to tune is λ_R, steering the risk-aversion.

In Figure 5, dependence of CAGR on λ_R (in Lasso General model) is depicted. It is possible to observe, that larger λ_R (more conservative portfolio) yields larger CAGR.

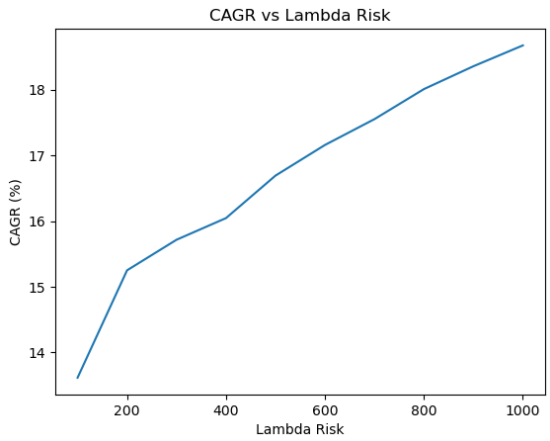

Figure 5: Dependence of portfolio CAGR on risk-aversion parameter (Lasso General model).

On top of that, in terms of CAGR dependence, such course was observed in all three analyzed model frameworks. Each portfolio, backed with predictions made by the particular model, has growing returns with growing value of λ_R.

Speaking of AV, unified course is observed as well. Figure 6 shows dependence of AV on the value of λ_R.

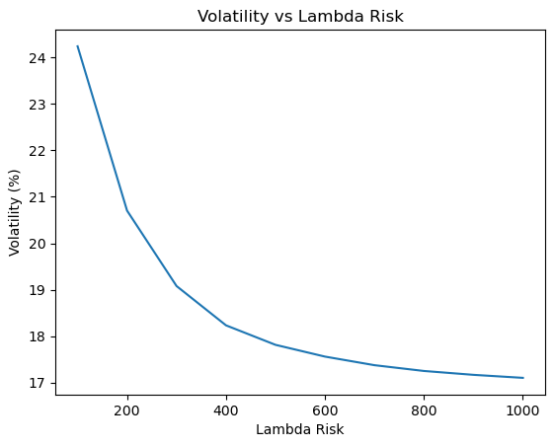

Figure 6: Dependence of portfolio AV on risk-aversion parameter (Lasso General model).

Even in this case, such declining course is observed in all three model frameworks. Even though course is universal for all the frameworks, each implementation was configured with of $\lambda_R = 1000$, as further growth has negligible effects on given evaluation metrics. In cases of Lasso and Ridge, General model variant was evaluated.

Evaluation results on the training-validation interval are overviewed in Table 4. Naïve strategy stands for portfoli optimization employed with naïve benchmark prediction model.

Portfolio Type	CAGR (%)	AV (%)	Sharpe ratio
Naïve	22.38	17.01	1.32
ARIMA	22.25	16.94	1.31
Ridge (SM)	21.68	16.89	1.28
Ridge (GM)	21.67	16.96	1.25
Lasso (SM)	18.58	17.08	1.09
Lasso (GM)	18.67	17.1	1.09

Table 4: Evaluation of tuned models on training-validation interval (SM – Stock-wise model; GM – General model).

As a benchmark was used scenario of investing capital into two indices S&P 500 and NASDAQ. Performance of these scenarios is overviewed in Table 5.

Index name	CAGR (%)	AV (%)	Sharpe ratio
S&P 500	7.69	21.75	0.35
NASDAQ	9.19	25.52	0.36

Table 5: Evaluation of benchmark models on training-validation interval.

Directly comparing results obtained by indices and models, portfolios backed by predictions perfomed marginally better in all reported metrics. On one hand, during the analyzed period, segment of IT experienced vast rise and this result is expected, but on the other hand, NASDAQ index consists mainly from the technology and IT companies, so one could expect similar performance compared to the model backed strategies [Mehle, 2023].

This means, that model backed strategies are capable of being both less risky and better in terms of average annual return.

Overview of ARIMA backed strategy and investments in S&P 500 and NASDAQ are visualized in Figure 7.

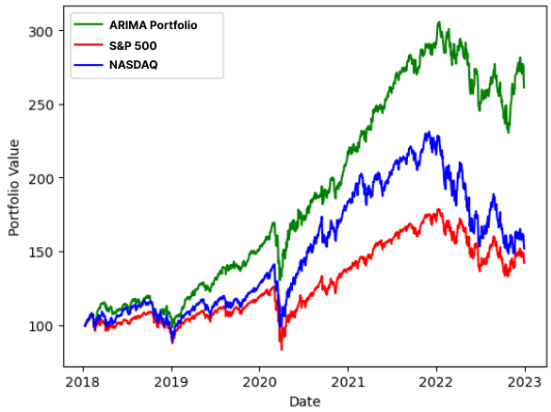

Figure 7: Overview of portfolio value across training-validation interval.

5.3 Test evaluation

Optimized portfolio strategies were applied on the test interval as well. Throughout results are given in Table 6 together with performance of benchmark stock indices in the same time period in Table 7.

Portfolio Type	CAGR (%)	AV (%)	Sharpe ratio
Naïve	17.2	26.35	0.65
ARIMA	16.47	26.14	0.63
Ridge (SM)	16.16	24.94	0.64
Ridge (GM)	15.98	24.95	0.64
Lasso (SM)	17.89	25.16	0.71
Lasso (GM)	17.64	25.19	0.7

Table 6: Evaluation of tuned models on test interval (SM – Stock-wise model; GM – General model).

Index name	CAGR (%)	AV (%)	Sharpe ratio
S&P 500	19.22	15.20	1.26
NASDAQ	26.11	20.34	1.28

Table 7: Evaluation of benchmark models on test interval.

Diametrally different evaluations of frameworks is observed. On the shorter test interval, benchmark indices overcame all the model backed strategies in all metrics. What is more is the opposite behavior of the models on the interval. While on the training-validation interval, models with the highest dominance reported the best results; on the test interval quite the opposite was observed.

Paradoxically, the framework with the worst prediction results delivered the best performing portfolio strategy. This is quite an interesting result in context of the market regime in the time interval. During this period, US market (particularly IT and technology segment) experienced **bull market** – phase in which market in general experiences higher returns and profits [Graeber, 2023].

This results suggest that MVP backed strategies performed poorly in strong bull market phase. On the other hand, in relatively mediocre phase between 2017 and 2022 prediction backed strategy performed above expectations.

6 Conclusions

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